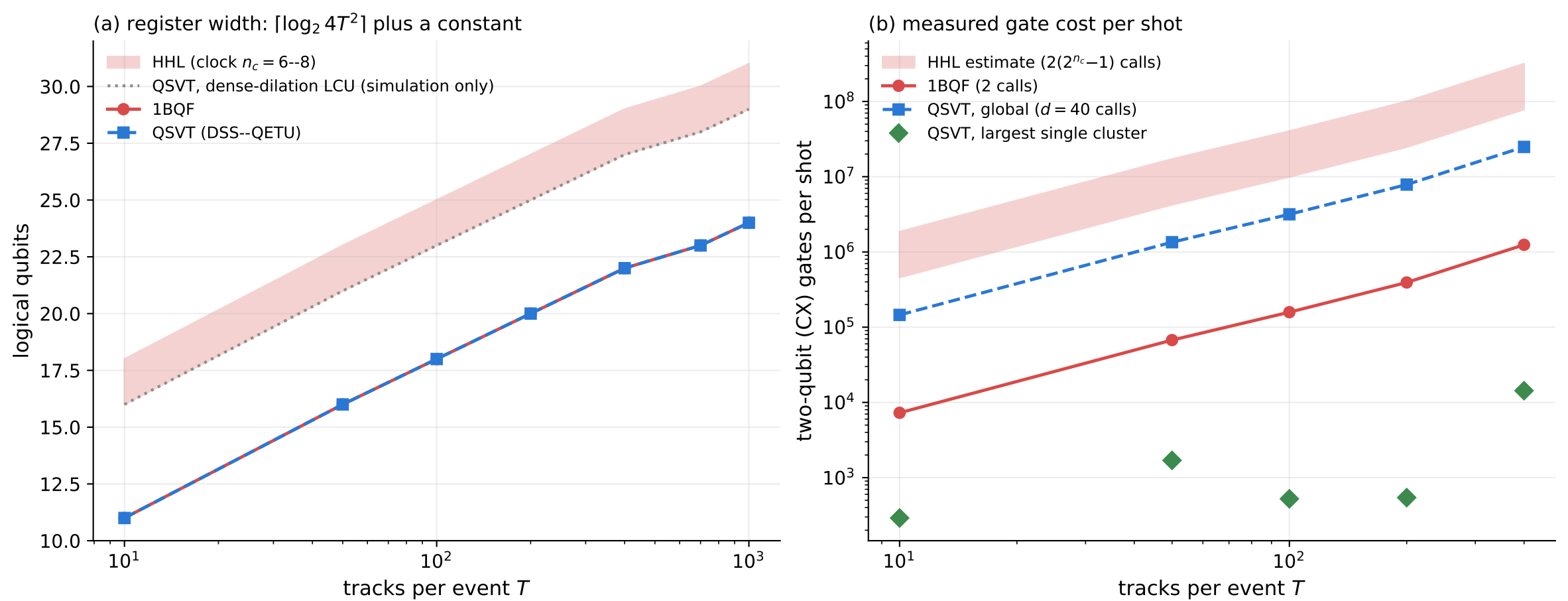
(a) register width: $\lceil \log_2 4T^2 \rceil$ plus a constant
HHL (clock $n_c = 6$--8)
QSVT, dense-dilation LCU (simulation only)
1BQF
QSVT (DSS--QETU)
logical qubits
30.0
27.5
25.0
22.5
20.0
17.5
15.0
12.5
10.0
10^1
10^2
10^3
tracks per event T
(b) measured gate cost per shot
HHL estimate ($2(2^{n_c}-1)$ calls)
1BQF (2 calls)
QSVT, global ($d = 40$ calls)
QSVT, largest single cluster
two-qubit (CX) gates per shot
10^8
10^7
10^6
10^5
10^4
10^3
10^1
10^2
tracks per event T